

Enantiomerically Enriched α -Borylzinc Reagents by Nickel-Catalyzed Carbozincation of Vinylboronic Esters

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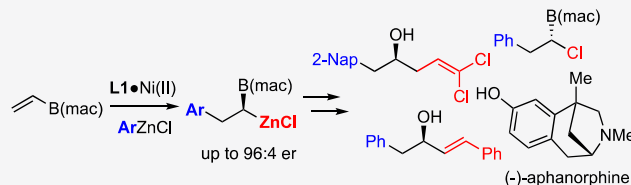


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ABSTRACT: In this paper is described a synthesis of enantiomerically enriched, configurationally stable organozinc reagents by catalytic enantioselective carbozincation of a vinylboronic ester. This process furnishes enantiomerically enriched α -borylzinc intermediates that are shown to undergo stereospecific reactions, producing enantioenriched secondary boronic ester products. The properties of the intermediate α -borylzinc reagent are probed and the synthetic utility of the products is demonstrated by application to the synthesis of (–)-aphanorphine and (–)-enterolactone.



INTRODUCTION

Since their first preparation in 1849,¹ organozinc species have become versatile organometallic reagents.² Because of their broad functional group compatibility, good configurational stability,³ and low toxicity, these reagents have been employed in a number of carbon–carbon and carbon–heteroatom bond forming reactions. While recent powerful reactions can employ racemic reagents for asymmetric synthesis,^{4,5} transition-metal catalysis has further broadened the utility of nonracemic organozinc reagents. The most general method for the synthesis of organozinc reagents is zinc insertion into organic halides; however, the mechanism of this process involves free radical intermediates,⁶ which renders asymmetric synthesis challenging. Alternative routes for the preparation of chiral organozinc reagents with high enantiomeric purity have been reported. Knochel developed an enantio- and diastereoselective hydroboration/boron–zinc exchange sequence (Figure 1a) and demonstrated that the resulting chiral organozinc reagents participate in a series of copper- or palladium-mediated stereospecific coupling reactions.⁷ More recently, Campos⁸ developed an enantioselective deprotonation/transmetalation method to prepare heterocycle-derived organozinc compounds (Figure 1b), and this strategy was later modified by Gawley to employ catalytic amounts of lithiated diaminoalkoxide (Figure 1c).⁹ Most recently, Knochel prepared chiral secondary organozinc reagents by stereoretentive lithium–halogen exchange followed by transmetalation (Figure 1d)¹⁰ and showed that the resulting dialkylzinc reagent undergoes stereoretentive cross-coupling with C(sp²) electrophiles. In terms of catalytic synthesis, carbozincation of π bonds, which represents a unique branch of carbometallation¹¹ reactions, provides an efficient route to organozinc reagents by the construction of carbon–carbon and carbon–zinc bonds in a single operation. Despite this intriguing reactivity, the carbozincation of alkenes has rarely been reported. Although zirconium(II),¹² iron(III),¹³ and tantalum-

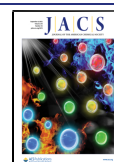
(IV)¹⁴ complexes catalyze intermolecular carbozincation of terminal alkenes, enantioselective additions to alkenes have only been achieved with strained and reactive double bonds, as seen in cyclopropanes (Figure 1e)¹⁵ and [3.2.1] oxabicyclic alkenes.^{11d} In this report, we describe a chiral Ni-based complex that facilitates the enantioselective carbozincation of vinylboronic esters and show that, under the appropriate reaction conditions, the α -borylzinc products are configurationally stable and may couple with a wide range of electrophiles to furnish chiral alkyl boronic esters (Figure 1f).¹⁶ The versatility of the resulting chiral boronic esters in stereospecific transformations¹⁷ offers a useful method for chemical synthesis.

RESULTS AND DISCUSSION

Our efforts in the development of alkenylboronate carbozincation were inspired during studies on a Ni-catalyzed tandem radical addition/cross-coupling¹⁸ where products consistent with carbozincation were observed with some systems. An initial optimization of this process revealed a nickel-L1 complex to be a prospective catalyst for the reaction of alkenylboronic ester **2** with phenylzinc chloride (**3**) in a THF/DMSO/DMF solvent mixture at –40 °C (Table 1, entry 1). Notably, the reaction yields are generally improved by the inclusion of an alkyl iodide (0.5–2 equiv; see the Supporting Information for detailed optimization studies), which was anticipated to facilitate the oxidation of low-valent inactive Ni species. After carbozincation, the α -borylzinc intermediate was

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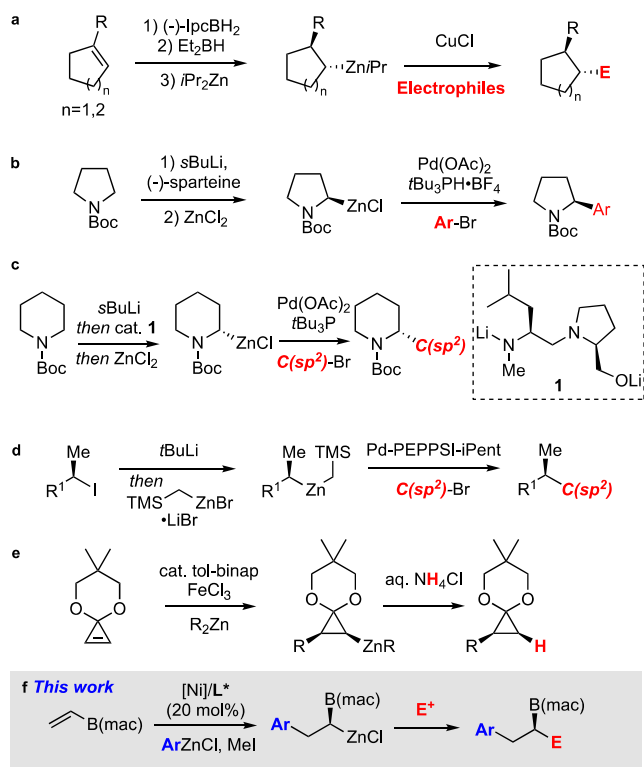
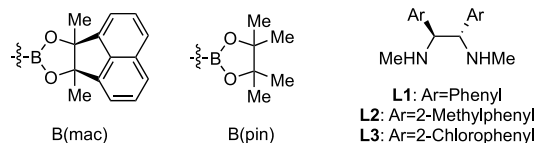


Figure 1. Synthesis of enantioenriched organozinc reagents: (a) hydroboration followed by boron–zinc exchange; (b) sparteine-promoted deprotonation; (c) catalytic enantioselective deprotonation; (d) stereoretentive lithium–halogen exchange followed by transmetalation; (e) carbozincation of cyclopropanes; (f) catalytic enantioselective carbozincation of a vinylboronic ester (this work).

Table 1. Effect of Reaction Parameters on the Ni-Catalyzed Enantioselective Carbozincation^a

entry	deviation from standard conditions	yield ^a (%)	er ^b
1	none	59	89:11
2	no nickel and L1	0	N/A
3	no nickel but with 20 mol % L1	0	N/A
4	no MeI	20	91:9
5	5/1 THF/DMSO as the solvent	63	86:14
6	5/1 THF/DMF as the solvent	13	65:35
7	B(pin) used instead of B(mac)	9	55:45
8	reaction conducted at −20 °C	57	80:20
9	L2 used instead of L1	42	91:9
10	L3 used instead of L2	40	95:5

^aYield was determined by ¹H NMR with an internal standard.



^bEnantiomeric ratio (er) was determined by chiral SFC analysis.

subjected to copper(I)-mediated allylation followed by oxidation of boronic ester, resulting in formation of the

homoallylic alcohol **4**. The high enantiomeric purity of **4** suggests that the intermediate α -borylzinc reagent not only is configurationally stable during the reaction but also reacts in a stereospecific fashion. Control experiments indicate that the nickel catalyst is indispensable for the reaction (entries 2 and 3) and that the use of methyl iodide (*vide infra*) and a THF/DMF/DMSO solvent blend is important for the highest yield and enantioselectivity (entries 4–6). Use of the pinacol-derived boronic ester vinylB(pin) in place of vinylB(mac) resulted in a diminished yield (entry 7; other alkenes such as styrene are unreactive). Of note, conducting the reaction at temperatures higher than −40 °C resulted in lower enantioselectivity (entry 8). Increasing the steric encumbrance of the ligand by installing substitution at the *ortho* position of the aryl ring increased the enantioselectivity but had a detrimental effect on the reactivity (entries 9 and 10).

With an effective reaction in hand, a selection of arylzinc reagents was surveyed in the two-step carbozincation/allylation sequence with allyl bromide serving as the electrophile. Aryl nucleophiles bearing alkyl and aryl groups at *meta* and *para* positions are competent reaction partners (Figure 2a; **5**, **6**, and **8**). Functional groups such as a ketal, alkene, and alkyne, as well as a TBS-protected alcohol, are tolerated (**7** and **9–11**). While aliphatic and alkenyl zinc reagents react with significantly diminished reactivity and/or selectivity, heterocycle (**12**)- and heteroatom-substituted (**13–17**) arylzinc reagents can also engage in the carbozincation process, providing the chiral alcohols with moderate to good yield and enantiomeric purity. A number of different electrophiles were also examined as trapping reagents for the carbozincation adduct. Allyl chlorides and bromides with a variety of functional groups and substitution patterns were subjected to the reaction and found to be effective (Figure 2b). Unprotected alcohols (**29**) and amines (**24**) can be tolerated and did not lead to competitive protonation of the C–Zn species. It was also shown that a conjugated enone (**26**) and enyne (**27**) can be generated with high diastereoselectivity. Synthetically useful structures such as an alkenyl halide (**28**), an alkenyl boronic ester (**30**), and an allyl acetate (**31**) can also be synthesized from readily available materials. In addition to allyl electrophiles, propargyl chloride is suitable for this reaction and furnishes homoallyl alcohol **32** with equally good yield and selectivity. Also of note, the α -borylzinc reagents can engage in stereospecific palladium-catalyzed Negishi cross-coupling with alkenyl (**33**) or aryl (**34**) iodides, preserving the enantiomeric enrichment of the boron-substituted stereogenic center. In this process, the use of lithium chloride as an additive was found to be crucial for preserving enantiomeric enrichment during the cross-coupling. Finally, enantiospecific bromination and chlorination (Br₂ and trichloroisocyanuric acid/CeCl₃, respectively) furnished enantiomerically enriched α -haloborates (**35** and **36**). That α -halo borates participate in stereoinvertive substitution reactions with carbon- and nitrogen-based nucleophiles provides a useful strategy for further elaboration of these motifs.¹⁹ Interestingly, the absolute configurations of the halogenation products (**35** and **36**) are opposite to those of the compounds obtained from the allylation and cross-coupling reactions.

The synthetic utility of the carbozincation products was probed by examining additional transformations and applications in natural product synthesis. In one experiment, it was shown that γ -butyrolactone **37**, a compound used by Hoveyda

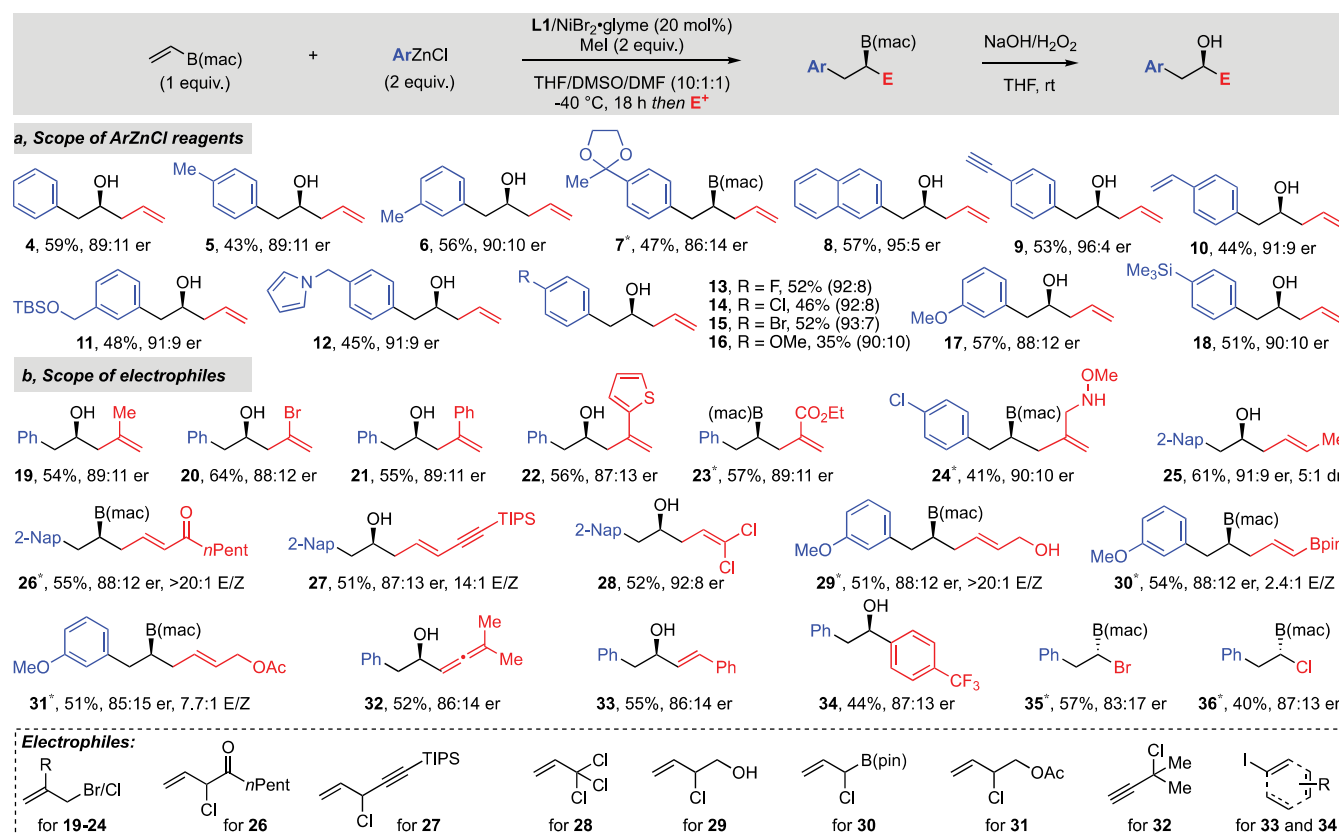


Figure 2. Scope of the catalytic enantioselective carbozincation reaction: (a) reactions with different ArZnCl reagents followed by trapping with allyl bromide; (b) reactions with $\text{C}(\text{sp}^2)\text{ZnCl}$ reagents and trapping with different electrophiles. Yields are of purified materials and are an average of two experiments (the remaining mass balance is generally unconverted starting material); er values were determined by chiral chromatography (HPLC or SFC) and have an error of $\pm 1\%$ (er values for compounds 24, 25, 29, and 35 have an error of $\pm 2\%$). Legend: (*) the compound was isolated as a boronic ester.

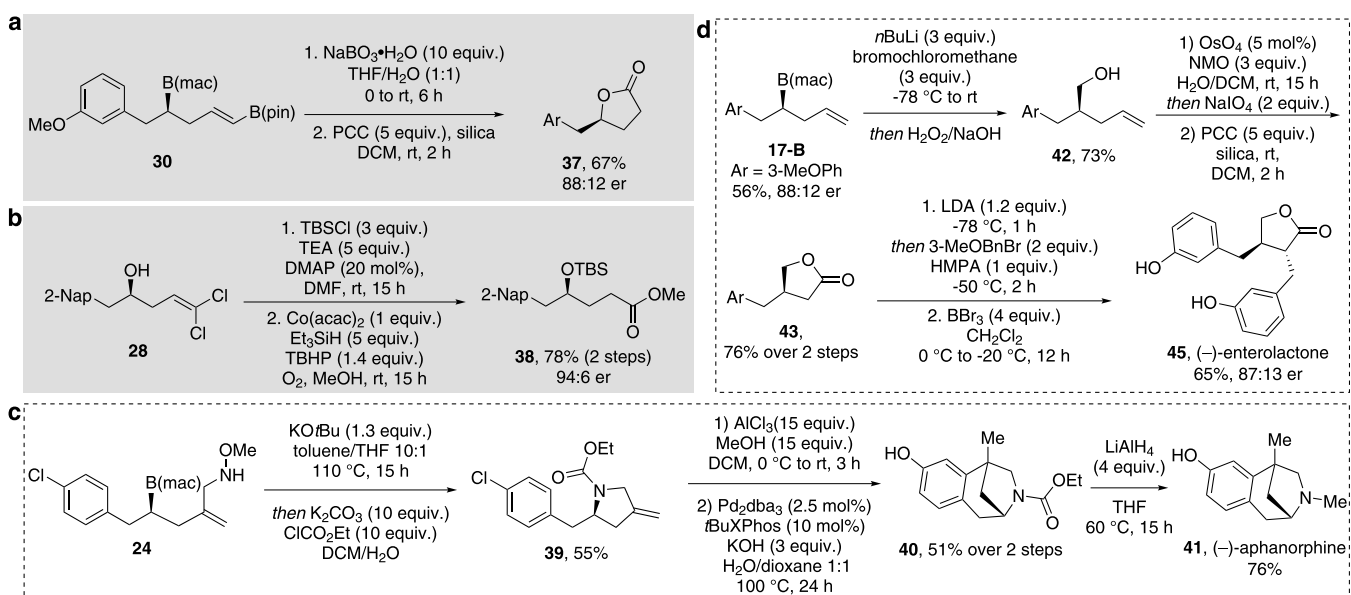


Figure 3. Transformations of the products: (a) synthesis of a γ -butyrolactone from 30; (b) synthesis of a γ -silyloxy ester by the oxidation of alkenyl dichloride 28; (c) synthesis of (–)-aphanorphine; (d) synthesis of (–)-enterolactone.

and co-workers²⁰ as an intermediate toward the synthesis of bruguierol A, could be accessed from bis(boronate) 30 by successive oxidation with sodium perborate and then PCC (Figure 3a). Also of note, while direct conjugate addition of

the α -borylzinc reagent has not yet been accomplished, allylation product 28 can be oxidized to an equivalent product, γ -silyloxy ester 38, by cobalt-mediated oxidation (Figure 3b).²¹ That the carbozincation/trap sequence can accommodate

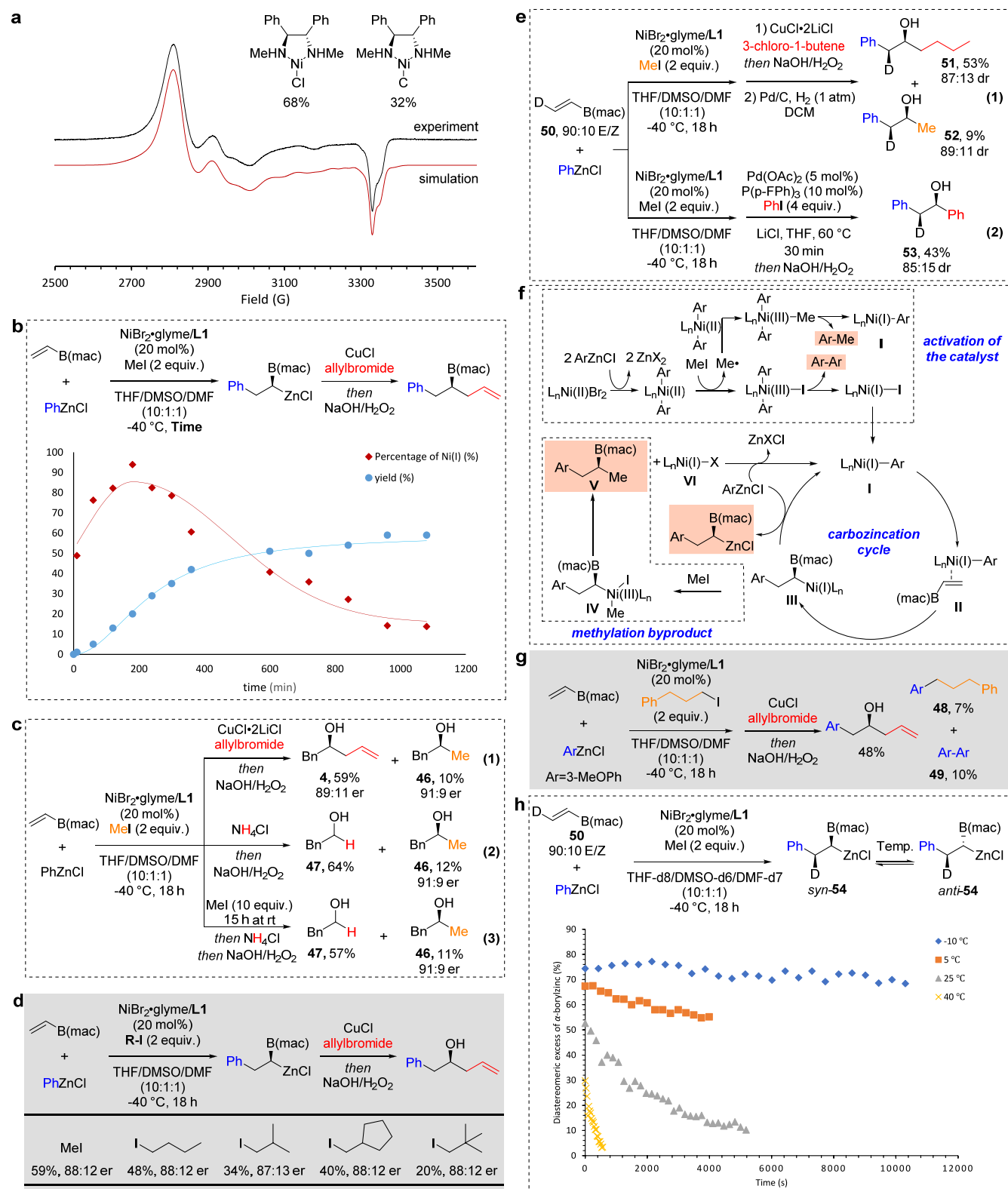


Figure 4. Mechanistic studies and proposed catalytic cycle: (a) X-band EPR spectrum of a reaction at 3 h (collected at 10 K with $\nu = 9.4$ GHz at 0.002 mW power and a modulation amplitude of 4 G) and simulated spectrum (parameters used: $g_1 = 2.002$, $g_2 = 2.379$, $g_3 = 2.269$, $A = 50.7$ G; $g_1 = 2.003$, $g_2 = 2.359$, $g_3 = 2.381$, $A = 26.4$ G); (b) percentage of Ni(I) species and reaction yield versus time; (c) control experiments to identify the origin of the methylation product; (d) carbozincation/trapping reactions employing *trans*-deuterium-labeled vinylB(mac); (e) influence of the increased steric hindrance of alkyl iodide on the yield and selectivity of the reaction; (f) proposed catalytic cycle; (g) isolation of the cross-coupling and homocoupling byproducts; (h) monitoring the epimerization of deuterium-labeled α -borylzinc at different temperatures by ^1H NMR.

unprotected methoxyamines allows the production of **24**, which was found to be a substrate for an intramolecular stereospecific amination²² reaction to give pyrrolidine derivative **39** (Figure 3c). In a short sequence, **39** could be converted to (–)-aphanorphine²³ (**40**) by Friedel–Crafts cyclization, palladium-catalyzed hydroxylation,²⁴ and LAH reduction. This new strategy currently provides the shortest reported route to this target. Finally, it was found that boronic ester **17-B** could be converted to chiral alcohol **42** through a Matteson homologation (Figure 3d), and this compound (**42**) serves as a convenient precursor to the lignan natural product (–)-enterolactone²⁵ (**45**) by oxidative lactonization, diastereoselective alkylation, and demethylation.²⁶

The intriguing reactivity described above prompted us to probe the reaction mechanism, and several observations pointed to the inner workings of the catalytic cycle. First, electron paramagnetic resonance (EPR) data were collected for a reaction under standard conditions (Figure 4a). A simulation of the spectrum suggests that it contains a mixture of an (L1)Ni^I-Cl complex ($g_{\perp} = 2.324$, $g_{\parallel} = 2.002$) and a (L1)Ni^I-C complex ($g_{\perp} = 2.370$, $g_{\parallel} = 2.004$) with the best fit to experimental data being a 68:32 ratio of these species. A comparison of the abundance of Ni(I) species with the reaction progress at multiple time points (Figure 4b) showed that the ratio of Ni(I) relative to other species is high during the period with the highest reaction rate (i.e., in the first 8 h), whereas a low percentage of Ni(I) species was observed when the reaction progressed at a lower rate. While this comparison does not necessarily rule out the activity of a small concentration of other highly reactive Ni species, it does suggest Ni(I) complexes as plausible catalysts. A second mechanistic clue was found in the formation of enantiomerically enriched methyl-containing compound **46**, which was obtained as a byproduct of the reaction (Figure 4c, eq 1). To learn about the origin of **46**, the carbozincation reaction was directly subjected to a protic workup, whereupon compound **46** was still obtained in addition to the primary alcohol **47** (eq 2); this observation indicates that the byproduct does not arise from copper(I)-mediated methylation. Of note, maintaining the carbozincation reaction for an extended time at elevated temperature and in the presence of excess methyl iodide did not increase the yield of **46** (eq 3), suggesting that **46** does not form by direct electrophilic trapping of the α -borylzinc reagent with methyl iodide. When primary alkyl iodides with increased steric hindrance were employed in place of methyl iodide, diminished reactivity was observed, yet the selectivity of the carbozincation reaction remained constant (Figure 4d). This result suggests that the alkyl group of the iodoalkane does not serve as a spectator ligand on the Ni complex during the stereochemistry-determining step of the process. Finally, when *trans*-deuterio vinyl boronic ester **50** was subjected to the standard conditions, net *syn* addition to the alkene was observed in both the carbozincation/allylation product (**51**) and in the methylation byproduct (**52**) (Figure 4e, eq 1), as well as in the carbozincation/Negishi coupling product (**53**). The stereochemical outcome of these processes suggests that the carbozincation occurs by concerted β -migratory insertion rather than aryl radical addition to vinylB(mac).

Collectively, the observations above may be accounted for by a catalytic cycle where carbozincation with an aryl-Ni(I) complex (I $\rightarrow$ II $\rightarrow$ III, Figure 4f) occurs and methylation byproduct **46** is generated by oxidative addition/reductive elimination involving an organonickel(III) intermediate such

as IV (Figure 4f). Following this scenario, the absolute configuration of the methyl-containing byproduct (**46**) likely reflects the configuration of the organonickel intermediate. Under this mechanism the carbozincation occurs in a redox-neutral fashion by olefin insertion and subsequent transmetalation with an arylzinc halide. The requisite generation of Ni(I) complexes from the NiBr₂ precatalyst can occur by a number of different mechanisms, but to our knowledge the generation of (diamine)Ni^I complexes has not been closely studied. Experimental work on related diphosphine,²⁷ bipyridine,²⁸ and pincer-based systems²⁹ suggests that mechanisms involving conversion of a fraction of Ni(II) to Ni(0), followed by comproportionation, can provide Ni(I) species. Alternatively, bimetallic oxidative addition may convert Ni(II) complexes to Ni(III) species and these, in turn, may be converted to Ni(I) by reductive elimination³⁰ (see dashed box, Figure 4f). To probe this possibility, 3-methoxyphenylzinc chloride and 3-phenylpropyl iodide were employed in a coupling reaction (Figure 4g). In addition to carbozincation products, cross-coupling product **48** and homocoupling product **49** were isolated in 10% and 7% yields, respectively. The formation of both of these compounds is consistent with bimetallic oxidative addition as a route to Ni(I), and given the marked positive impact of alkyl iodides on the carbozincation reaction, we expect that the bimetallic oxidative addition is the more likely path to catalyst activation.

The ability of the enantioselective carbozincation to provide a useful route to nonracemic chiral building blocks, outside of the transformations presented in the examples above, will rely on an understanding of the configurational stability of the α -borylzinc intermediate. To learn more about this feature of the α -borylzinc compounds, carbozincation of deuterium-labeled vinylB(mac) **50** was conducted and the configurational stability of the labeled α -borylzinc compound **54** was monitored by ¹H NMR at different temperatures. As shown in Figure 4h, the diastereomeric excess of **54** remained constant when the compound was maintained at –10 °C for 3 h. Epimerization is more rapid at 5 °C, and at 40 °C complete scrambling occurs within 10 min. Thus, effective use of these reagents in asymmetric synthesis appears to require rigorous maintenance of low-temperature reaction conditions; however, we note that the addition of LiCl allows stereospecificity in Negishi couplings at higher temperatures and expect that this additive may be of use in other contexts.

CONCLUSION

In summary, a catalytic enantioselective carbozincation of vinylboronic esters furnishes enantioenriched α -borylzinc reagents. The chiral α -borylzinc reagents can be directly converted to chiral boronic esters by copper(I)-mediated or palladium(II)-catalyzed coupling reactions or by halogenation reactions in a stereospecific fashion. Mechanistic studies with a deuterium-labeled substrate and EPR experiments revealed a plausible catalytic cycle involving a nickel(I) complex that catalyzes the carbozincation reaction in a redox-neutral fashion. Efforts to extend this process to other alkenes and other organozinc reagents are ongoing.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.1c05274>.

Procedures, characterization, and spectral data (PDF)

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Notes

The authors declare no competing financial interest.

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